Bitcoin Market Monitor

Data, Method, and Design



University of Massachusetts Amherst
By: Aiyman Akbar
Date: 10/16/2026
Dashboard Link

1. The purpose of this memo

This memo accompanies the Bitcoin Market Monitor, a public dashboard I built that compares the most recent hours of Bitcoin trading against the asset's own history. It explains the question the dashboard helps a reader explore, the data and method behind it, what the results do and do not show, and the choices I made in presenting them.

I chose to build the story around one deliberately narrow question. When has Bitcoin looked like this before, and what tended to follow? The dashboard does not forecast prices and is not a trading tool. It is a way of placing the current moment in historical context, with the uncertainty of that context shown rather than hidden.

2. Data, method, and what the dashboard shows

2.1 Starting with the simpler view

The dashboard opens with a live price ticker and three charts that make sense to a reader with no statistical background. The first plots daily closing prices from September 17, 2014 to the present on a logarithmic axis, so that the early years, when Bitcoin traded in the hundreds of dollars, and the recent years, when it has traded above $100,000, are both legible. The second is a histogram of daily percentage moves. The third shows the most recent 90 days.

These charts establish three facts that the rest of the dashboard depends on. Bitcoin has moved through repeated booms and crashes rather than along a steady trend. Its daily moves are large by the standards of most assets, and I wanted that to be easy to see. Across the daily history, the median absolute daily move is about 1.4 percent, about 12 percent of days move more than 5 percent in either direction, and the largest drop in a single day was 37 percent on March 12, 2020. The most recent 90 days, whatever they contain at the time of viewing, are a small slice of that history. Together these charts give a non technical reader the background to make sense of the pattern matching that follows.

2.2 Data and refresh

The dashboard uses two price series. Daily candles run from September 17, 2014 to the present and were originally sourced from Yahoo Finance. Hourly candles run from May 15, 2018 to the present. The hourly history was backfilled from a free Bitstamp BTC/USD archive, which began in May 2018 and therefore sets the floor for hourly coverage. Going further back would have required a paid data source which I did not pursue for this project.

A scheduled job on GitHub Actions refreshes the data every 12 hours, at 00:00 and 12:00 UTC. Since September 2026 the job reads new candles from Binance's public market data mirror, with Yahoo Finance as a fallback. The original design used Yahoo Finance first, and I reordered the sources after the scheduled job kept failing because the data providers rejected requests coming from Github's servers. Each run appends the new candles, rebuilds the features, reruns the pattern matching for every window, and commits the results, so the dashboard stays current without manual intervention. The "as of" timestamp on the page is the time of the last hourly candle in the data.

2.3 How the pattern matching works

Each hour, or each day for the one week window, is described by nineteen features computed from price and volume. I put them into groups so they would be easier to display below.

- Price level and activity: the closing price and the trading volume.
- Recent returns: the one period return and the returns one, two, and six periods earlier.
- Trend and volatility: rolling means and rolling standard deviations of price over 6 and 24 periods.
- Momentum: 12 and 26 period exponential moving averages, their difference (MACD), and a 14 period relative strength index.
- Position relative to trend: the distance from the 24 period moving average, in standard deviation units.
- Volume dynamics: the one period change in volume and a 24 period volume z score.
- Time: the hour of day and the day of week.

Each feature is then standardized with a rolling z score over a 30 day window. At every point in time, the value is expressed as a number of standard deviations from its own trailing 30 day mean. This step is what makes 2018 and 2026 comparable. I wanted the comparison to be about how the market was behaving relative to its own recent norm rather than about dollar prices, which is why I chose this method.

I defined a window as the stack of standardized features over the chosen lookback. For the three shorter options that means 12, 24, or 72 hours of hourly data. For the one week option it means 7 days of daily data. The current window is compared against every past window of the same length using cosine distance, which measures the angle between two feature vectors. I chose cosine distance because it responds to the shape of a pattern and largely ignores its scale, and shape was what I wanted to compare. The ten past windows with the smallest distance become the matches. I thought ten was enough to show a spread of outcomes while still fitting in a table a reader can go through one row at a time. I only allow a past window to be a match if enough history follows it to measure every forward horizon, so every match comes with a complete set of outcomes.

For each match, the pipeline records what Bitcoin did next as the percentage change in the actual closing price from the end of the matched window. I scaled the forward horizons with the lookback so the follow up period stays proportionate to the pattern being matched. The 12 hour window looks ahead 1, 6, 12, and 24 hours. The 24 hour window looks ahead 1, 12, 24, and 72 hours. The 72 hour window looks ahead 6, 24, 72, and 168 hours. The one week window looks ahead 1, 3, 7, and 30 days.

2.4 What the dashboard reports

I chose to summarize the results as a range rather than a single number, because a single number would claim more than the method can support. The explanation panel states how many matches were found, the dates they span, and the median forward return at the longest horizon for that window together with the 10th to 90th percentile band across the ten matches. On October 15, 2026, for example, the 24 hour view reported ten matches between June 2018 and January 2025. Three day returns after those matches ranged from minus 8.6 percent to plus 2.3 percent with a median of minus 1.0 percent. The width of that band is what I want a reader to notice first. When the matches disagree, history offers little guidance, and I wanted the dashboard to show that disagreement instead of averaging it away.

Two views sit below the summary. The pattern overlay plots the current window in orange against the ten matches in gray. I rescaled every line to start at 0 percent so that shapes can be compared without the price level getting in the way, since a 2018 window and a 2026 window will never share a dollar scale. Hovering over a gray line reveals its date. The historical similarity table lists the same ten matches with their start date, their distance score, and their forward return at each horizon, for readers who want to see the individual cases behind the summary. A lookback selector switches between the four windows, and the whole section updates.

2.5 A calculation error and how it was corrected

Early in development I found that the forward returns were being computed from the standardized closing prices rather than the raw ones. Because z scores can be negative or close to zero, dividing one by another produces meaningless ratios. The output reported a gain of about 200 percent over 24 hours for a window where the actual move was a loss of under 3 percent. I caught this before the dashboard went live. The fix carries the raw closing price through the pipeline as a separate column and computes forward returns in price space. I also added two regression tests so the error cannot come back quietly. One bounds every forward return within plus or minus 50 percent on a synthetic price series where the true moves are small. The other recomputes a forward return directly from the raw prices and checks it against the pipeline's output. I include this because it is the kind of error that would have undermined every number on the dashboard, and it is why I now check results by hand against the raw data before trusting them.

2.6 Testing whether the matches carry information

Finding similar windows is not the same as finding useful ones, and I did not want to rely on a disclaimer to make that point. I ran a walk forward test on the matcher exactly as it is deployed. For every day since May 2020 for the hourly windows, and since November 2016 for the one week window, the test rebuilt the ten matches using only the history available on that day, recorded their median and their 10th to 90th percentile band, and compared both with the return that actually followed. As a baseline I had it do the same with ten randomly chosen past windows, repeated 200 times per day. I decided on the two headline measures before running anything so I could not pick the flattering ones afterwards. They were how often the band contained the realized return and how often the median had the right sign, both on the default view of a 24 hour window and a 72 hour horizon.

The matches did no better than the random windows. Across 2,338 days, the band contained the realized return 67 percent of the time. For random windows it was 72 percent. The median had the right sign 51 percent of the time. For random windows it was 50 percent, and always guessing up would have scored 53 percent. The rank correlation between the match median and the realized return was 0.04. The other fifteen combinations of window and horizon look the same, with hit rates between 50 and 54 percent and correlations near zero.

I see this as the most important result in the memo. The matches describe the past. They do not help with what comes next, and that is the strongest reason I have for presenting the dashboard as context rather than as a forecast.

2.7 Limits and how to read the results

I built the dashboard as historical pattern recognition, and I want its results read as context, not as a forecast. Here are some of the limitations I think matter the most.

- Similar past windows do not guarantee similar outcomes. Ten matches is a small sample, and the spread of their outcomes is usually wide.
- Market conditions change. The hourly history spans halvings, the approval of spot Bitcoin ETFs, changes in regulation, and shifts in who trades the asset. A match from 2018 may not generalize to current conditions, and I did not weight recent matches more heavily than old ones.
- The matches are not independent. Nothing in the method prevents two matches from overlapping. On October 15, 2026, for instance, the 72 hour view included two windows that began one hour apart in August 2018, so their forward returns are close to the same observation counted twice.
- Similarity across nineteen features is not the same as similarity in whatever drives the next move. The method does not test for causation and I do not claim it.
- The hourly history is stitched together from three sources whose volume figures sit on different scales. The two volume features are relative measures within a rolling window, so the distortion is confined to the hours around each seam, but I know it is there.
- The walk forward test measured the method as I built it. It did not try other feature sets, distance measures, or numbers of matches, so it says this version has no edge, not that no version could.

For these reasons the dashboard reports ranges rather than point estimates, and is described as an educational tool, not a predictive financial model. It is not investment advice, and I did not build it to be used that way.

3. My thinking behind the final structure

For the structure, I wanted to tell a data-driven story, so I placed the simpler Tableau visuals at the start to get people familiar with Bitcoin data, what it looked like over time, and what it looks like day to day. From there I moved on to the more complicated pattern recognition window, so that by the time readers reach it they have the background to interpret it.

I chose a carousel format so there would be less scrolling and a cleaner page. For color, I wanted the same scheme as the Bitcoin logo, or the one Bitcoin had in my mind, which was dark gray, orange, and a little white. I also included text throughout the website so it would be a combination of visuals and writing, which I felt would be best for people with less experience in finance or cryptocurrency.

I kept the language as simple as I could and explained any jargon I used, because cryptocurrency does come with some basic jargon. I was also inspired by the layout of the Microsoft 365 dashboard page, and I tried to make mine clean and easy on the eyes in the same way.

4. Building it and what's next in my mind

I did not use any generative AI to come up with the storytelling or to write anything for me. I did use Claude Code as a coding assistant for the more technical work, like scaffolding the Next.js app and helping me set up the Python pipeline. It also helped when I was wiring the embed links for the Tableau visuals together, since they all use the same link. I also wanted to try something new for the front end to make it look good, so I used the new Claude design tool, where I gave it screenshots of my existing dashboard and described what I wanted it to look like. It gave me back a set of files that I integrated into a branch in my repository and then merged into main at the end.

Something I want to do in the future is add other digital assets, maybe Ethereum or Solana, and possibly some stocks, not only cryptocurrency. I would also like to add a longer window pattern, maybe multi-month, as the daily history accumulates.